

Sean Luka Girgis

Data Engineer | SQL, Python, ELT, Data Quality & Warehouse Reporting

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Professional Summary

Senior Data Engineer with 20+ years of enterprise technology experience across SQL, Python, ELT/ETL pipelines, data warehousing, data modeling, data quality, validation, documentation, reporting/dashboard support, Git concepts, and cloud data platforms. Strong production background with Oracle, Amazon Redshift, AWS S3/Glue/Athena, PySpark, and Python-driven automation, plus experience building trusted analytics datasets and validating reporting outputs in regulated enterprise environments. Best aligned to SQL/Python data engineering roles requiring warehouse transformations, quality checks, documentation, Git-based modular development, reporting support, and stakeholder communication; Snowflake, dbt, OpenFlow, healthcare/EMR systems, HIPAA-specific workflows, Cognos/SAS/SPSS/Crystal Reports, and Microsoft SQL Server are positioned carefully as adjacent or learning areas rather than overstated production ownership.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built SQL, Python/Pandas, and PySpark ETL/ELT pipelines ingesting telemetry from 6,000+ infrastructure endpoints and transforming raw operational data into curated datasets for reporting, analytics, forecasting, validation, and decision support.
- Designed AWS and warehouse-backed data platform components using S3, AWS Glue, Athena, Amazon Redshift, Oracle, SQL, and PySpark to support scalable ingestion, transformation, querying, and downstream analytics.
- Developed SQL-heavy transformation workflows, reporting tables, data marts, validation queries, and reusable datasets for technical and business stakeholders in a regulated financial-services environment.
- Implemented data quality and reconciliation practices across BMC TrueSight/TSCO, CA Wily/APM, AppDynamics, Oracle, and AWS-backed reporting layers to improve accuracy, completeness, consistency, and trust in reporting outputs.
- Optimized complex SQL queries, Oracle schemas, Redshift workloads, indexes, views, partitioning patterns, and reporting data models for performance, maintainability, and cost-aware processing.
- Documented data flows, source mappings, metric definitions, transformation rules, dependencies, validation checkpoints, and support procedures to improve visibility and reuse across reporting workflows.
- Used Python scripting and SQL to automate data manipulation, validation, reconciliation, reporting preparation, and production troubleshooting.
- Partnered with infrastructure, analytics, application, engineering, and business teams to translate requirements into reliable datasets, dashboards, documentation, and operational recommendations.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Built analytics and reporting outputs from Dynatrace AppMon and Gomez Synthetic Monitoring data for Brand.com and critical hospitality systems.

- Investigated real-user and synthetic monitoring source data to identify data quality issues, performance trends, and optimization opportunities.
- Created dashboards and reporting views that converted raw monitoring data into actionable performance, reliability, and stakeholder insights.
- Collaborated with application and infrastructure teams to validate findings, document issues, and support practical performance improvements.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Served as CA APM/Introscope SME for a large financial-services environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Built Perl and KornShell automation for extracting, validating, transforming, and distributing APM telemetry, replacing manual reporting workflows with repeatable data-processing pipelines.
- Designed dashboards, alert rules, SLA thresholds, reporting standards, and technical documentation used by application, middleware, infrastructure, and operations teams.
- Provided troubleshooting guidance, onboarding support, training material, and knowledge sharing to improve adoption of monitoring and reporting standards.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Developed and executed performance test plans for enterprise telecom applications, including baseline comparisons, regression review, and release-readiness analysis.
- Analyzed application behavior under load to identify throughput limits, SQL/JDBC bottlenecks, CPU, memory, thread, and garbage-collection issues.
- Documented test results, expected behavior, defects, risks, and remediation recommendations for engineering and release teams.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster, preserving transaction performance and data integrity.
- Built validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Automated schema conversion, data validation, phased cutover support, troubleshooting, and production-readiness activities for a mission-critical migration.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science

Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering

Zagazig University, Egypt

Skills

Flagship Projects

Serverless Lakehouse Platform (AWS)

Designed a scalable AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet for raw ingestion, transformation, curated analytics datasets, query optimization, reporting access, and validation checkpoints.

Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, Parquet, SQL

Production Data Quality and RCA Workflow

Built validation and reconciliation workflows for production reporting datasets, including row-count checks, null checks, missing feed diagnosis, unmatched join checks, and issue documentation.

Technologies: SQL, Python, Pandas, Data Validation, RCA, Runbooks

HorizonScale — Capacity Forecasting Engine

Built Python/PySpark workflows to process infrastructure telemetry, validate pipeline outputs, prepare forecasting datasets, and deliver capacity insights through repeatable reporting workflows.

Technologies: Python, Pandas, PySpark, SQL, Prophet, Data Validation